Bryan Torres

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech) Blacksburg, VA
Master of Engineering in Computer Science & Applications Expected May 2027
Graduate Coursework GPA: 3.90/4.00 | Focus: Computer Systems & Architecture, Software Systems
Bachelor of Science in Computer Science May 2026

EXPERIENCE

Software Engineer Intern | Expedition Technology May 2026 -- Aug. 2026
Geospatial Dataset Diversity *Herndon, VA*

- Built a FastAPI endpoint that scores the diversity of geospatial training datasets, and presented a **company-wide live demo** using mock data.
- Calculated **normalized Shannon entropy** across geography, time, lighting, and image quality, and built TypeScript charts and maps that explain each score.
- Automated Kubernetes database migrations with reusable Helm and Argo CD templates, ordering deployment steps so a **failed migration blocks the release**.

Software & DevOps Engineer Intern | Dark Wolf Solutions Jun. 2025 -- Aug. 2025
Project Syndicate *Herndon, VA*

- Co-developed a **Next.js dashboard** that combines pipeline health, security, and compliance data, with views by application and by organization.
- Deployed four Dockerized services to Google Kubernetes Engine with Argo CD and Kubernetes Ingress, and reduced Docker image size by **22%**.
- Set up GitLab CI/CD security gates with SonarQube and OWASP checks, and wrote Jest/pytest tests that reached **80% coverage**.

Undergraduate Researcher | Virginia Tech Oct. 2024 -- Jan. 2026
MARCO -- Multi-Agent Code Optimization for HPC *Blacksburg, VA*

- Extended MARCO to turn baseline code into **CUTLASS CUDA kernels**, using library guides and automated testing to catch and repair generation errors.
- Compared step-by-step guides with open-ended CUTLASS exploration, finding **lower API costs with guides** for the same kernel generation task on V100 GPUs.
- Co-authored *MARCO* (arXiv:2505.03906, 2025).

PROJECTS

Codeward | Rust Cryptographic Core (In Development) Oct. 2026 -- Present
Rust, Argon2id, HKDF-SHA256, XChaCha20-Poly1305, RustCrypto

- Built the **Rust core** for an offline password manager, supporting encrypted vault creation, credential updates, and master-password recovery with a paper recovery code.
- Prototyped **macOS vault storage** with file locking and atomic file replacement, testing interrupted saves with simulated storage failures and forced process termination.
- Organized separate **session and save components**, keeping decrypted credentials out of the save worker so the vault can lock without waiting for disk writes.

Kingdom Tycoon | Roblox Game (In Development) Jun. 2026 -- Present
Luau, Roblox Studio, DataStoreService, ProfileStore

- Organized a **typed Luau backend** into separate server services and client controllers, each with an explicit startup sequence.
- Built player-profile saving with **session locking**, automatic saves, and revision checks that reject invalid or stale updates.

TECHNICAL SKILLS

Languages: Rust, Python, C, TypeScript, JavaScript, SQL, Java, Luau
Infrastructure & Cloud: Linux, Docker, Kubernetes, GKE, Helm, Argo CD, GitLab CI/CD, Git
Frameworks & Data: FastAPI, React, Next.js, PostgreSQL, SQLAlchemy, Alembic, Chart.js
Testing & HPC: pytest, Jest, CUDA, PyTorch, Hugging Face Transformers, Slurm, NVIDIA Nsight